

1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a * b = [a_1, \dots, a_n, b_1, \dots, b_m]$, where $a = [a_1, \dots, a_n]$ and $b = [b_1, \dots, b_m]$.

$$\begin{aligned} \mathfrak{E} &:= \mathfrak{T} \cup \{\mathfrak{e}_1 * \mathfrak{o} * \mathfrak{e}_2 \mid \mathfrak{o} \in \{[“+”], [“-”]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\ \mathfrak{T} &:= \mathfrak{F} \cup \{\mathfrak{t}_1 * \mathfrak{o} * \mathfrak{t}_2 \mid \mathfrak{o} \in \{[“\times”], [“\div”]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\ \mathfrak{F} &:= \mathfrak{N} \cup \{[“(”] * \mathfrak{e} * [“)”] \mid \mathfrak{e} \in \mathfrak{E}\} \\ \mathfrak{N} &:= N \cup N^* \\ N &:= \{“0”, \dots, “9”\} \end{aligned}$$

What is the difference between pattern matching and tokenwise parsing?

2 gen.jl and pattern.jl

The *map form* of a pattern $\mathfrak{P}$ is a function $\Phi' : \mathfrak{P} \rightarrow L$, where $L \subseteq \mathcal{P}(\mathfrak{P})$ is a set of *labels*¹. The *inverse map form* of a pattern $\mathfrak{P}$ is a function $\Phi : L \rightarrow \mathcal{P}(\mathfrak{P})$, where $\Phi(l) = \{p \in \mathfrak{P} \mid \Phi'(p) = l\}$ for $l \in L$.

Let $G := (\langle \mathcal{G} \rangle, *)$. Let $\Delta : [\Phi, L^2] \rightarrow \Phi$, where Φ is the inverse map form of a pattern $\mathfrak{P} \subseteq G^{<\omega}$ and $L = \text{Dom}(\Phi) \subseteq G$, be defined by $\forall l \in \text{Dom}(\Phi_1) \ (\Phi_2(l) = \Phi_1(l))$ and $[l_1, l_2] \in \Phi_2(l_1 * l_2)$ where $\Phi_2 = \Delta(\Phi_1, [l_1, l_2])$ and $l_1, l_2 \in \text{Dom}(\Phi_1)$.

¹Labels – or, *concepts* – are useful abstractions of patterns, which will be used later to optimize the complexity of solution-finding systems. The subsets $\mathfrak{T}, \mathfrak{F}, \mathfrak{N}$ and N in example 1.1.1 are examples of labels.